

# LAB 3 REPORT

## Component Modelling & Architectural Pattern Selection

### Student Club Event Ticketing & Budget Portal

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**Course: Software Engineering**

## 1. Objective

The objective of this lab is to evaluate suitable architectural styles for the Student Club Event Ticketing & Budget Portal, select the most appropriate architecture, and model the system using a UML Component Diagram showing components, interfaces, dependencies, and interactions.

## 2. Scenario Review

The Student Club Event Ticketing & Budget Portal is a web-based system that supports event proposal submission, budget approval, QR-code ticket generation, ticket booking, registration monitoring, and role-based authentication.

Primary actors:

- Club Lead — submits event proposals, requests budgets, generates tickets after approval, and monitors registrations.
- Campus Admin — reviews proposals and approves or rejects proposed budgets with remarks.
- Student — browses approved events, purchases tickets, and views/downloads QR-code tickets.

## 3. Requirements Identified

| Type        | Requirement                                                  | Priority |
|-------------|--------------------------------------------------------------|----------|
| Functional  | Submit event proposal and budget                             | High     |
| Functional  | Review and approve/reject proposal                           | High     |
| Functional  | Generate unique QR-code tickets                              | High     |
| Functional  | Purchase tickets after successful payment                    | High     |
| Functional  | View/download QR tickets                                     | Medium   |
| Functional  | Monitor event registrations                                  | Medium   |
| Security    | Authenticated and role-authorized access                     | High     |
| Performance | 95% of requests respond within 3 seconds under expected load | Medium   |

## 4. Architectural Style Analysis

| Architecture  | Structure                                                  | Advantages                                                     | Limitations                                                                                           |
|---------------|------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Layered       | Horizontal layers such as presentation, business and data. | Clear separation of concerns; easy to understand and maintain. | Potential performance overhead; tighter coupling between layers; individual scaling is less flexible. |
| Microservices | Small, independently                                       | Independent scaling,                                           | Operational                                                                                           |

|               |                                                         |                                                                        |                                                                     |
|---------------|---------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------|
|               | deployable services.                                    | fault isolation and independent deployment.                            | complexity, network latency and data consistency challenges.        |
| Client-Server | Multiple clients communicate with a centralized server. | Centralized control, simple deployment and consistent data management. | Possible single point of failure and server scalability bottleneck. |

## 5. Architecture Selection

### Selected Architecture: Layered Architecture

Layered Architecture was selected because the portal has clearly separated responsibilities for user interaction, event and budget processing, ticketing, payment, authentication, registration, and data management. The system is also a centralized university portal where the operational complexity of a microservices architecture would not provide a proportional benefit.

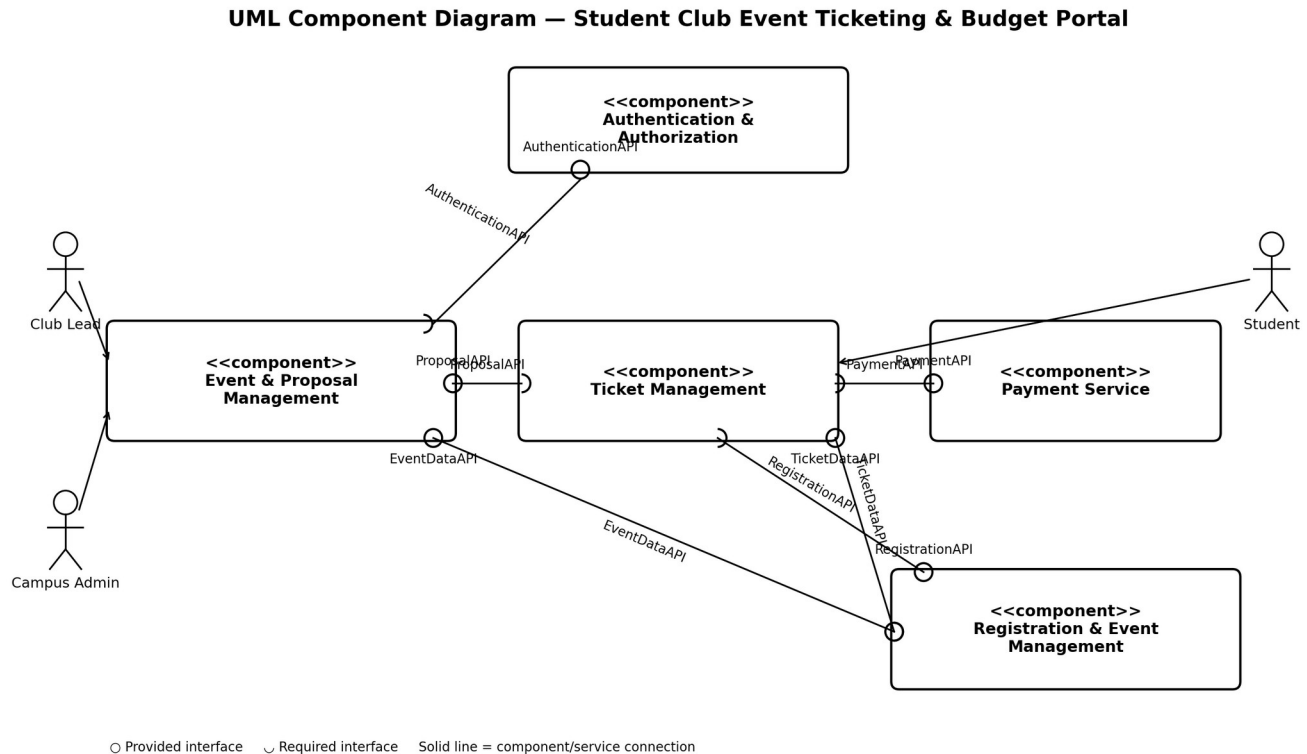
1. Separation of responsibilities: presentation, business logic and data operations can be organized into distinct layers and components.
2. Centralized consistency: event, budget, payment, ticket and registration records can be managed through controlled application logic.
3. Security: authentication and role-based authorization can be enforced before protected business operations are executed.
4. Performance: frequently used operations such as approved-event retrieval, ticket availability checks and ticket access can be handled within the appropriate application components.

## 6. Component Identification

| # | Component                       | Responsibility                                                                        |
|---|---------------------------------|---------------------------------------------------------------------------------------|
| 1 | Authentication & Authorization  | User login, credential validation, role identification and role-based access control. |
| 2 | Event & Proposal Management     | Event proposal submission, budget requests, proposal review and status management.    |
| 3 | Ticket Management               | Ticket generation, unique ticket IDs, QR-code creation and ticket availability.       |
| 4 | Payment Service                 | Payment requests, payment processing and payment confirmation.                        |
| 5 | Registration & Event Management | Participant registrations, ticket availability and registration monitoring.           |

## 7. UML Component Diagram

The component diagram below models the five major components and their service interfaces. Provided interfaces are represented using ball notation and required interfaces using socket notation.



## 8. Interface Specification

| Interface         | Provider                        | Consumer                        | Purpose                                              |
|-------------------|---------------------------------|---------------------------------|------------------------------------------------------|
| AuthenticationAPI | Authentication & Authorization  | Event & Proposal Management     | Authentication and authorization verification.       |
| ProposalAPI       | Event & Proposal Management     | Ticket Management               | Approved event/proposal information.                 |
| PaymentAPI        | Payment Service                 | Ticket Management               | Ticket payment processing requests and confirmation. |
| RegistrationAPI   | Registration & Event Management | Ticket Management               | Registration and ticket availability information.    |
| EventDataAPI      | Event & Proposal Management     | Registration & Event Management | Approved event and event details.                    |
| TicketDataAPI     | Ticket Management               | Registration & Event Management | Ticket records and availability.                     |

## 9. Written Architectural Justification

**Architecture Selection:** We chose Layered Architecture for the Student Club Event Ticketing & Budget Portal.

**Architectural Choice:** The Layered Architecture is appropriate because the portal contains clearly separable responsibilities such as user interaction, event and budget processing, ticket management, payment processing, authentication and registration. These responsibilities can be organized into controlled layers and components.

**Reason 1:** The architecture provides clear separation of responsibilities. Proposal submission, budget approval, ticket generation, payment processing and registration management can be handled through dedicated components, improving maintainability.

**Reason 2:** The portal requires consistent management of event proposals, budgets, ticket availability, payments and registrations. A centralized business layer provides controlled access to the data layer and helps maintain consistent records.

**Security Advantage:** A dedicated Authentication & Authorization component can enforce authentication and role-based access control. Students, Club Leads and Campus Admins receive access only to functions appropriate to their roles.

**Performance Benefit:** Separating responsibilities allows frequently accessed operations such as approved-event retrieval, ticket availability checks and ticket viewing to be handled efficiently within their appropriate components, supporting the three-second response requirement under expected load.

## 10. Conclusion

The Lab 3 analysis identified Layered Architecture as the most suitable architectural style for the Student Club Event Ticketing & Budget Portal. Five major components and six service interfaces were identified to represent authentication, event proposal management, ticketing, payment, and registration functionality. The resulting component model provides clear separation of responsibilities, supports role-based security, and provides a structured basis for implementation.

## 11. Jira Agile Backlog Integration

The project requirements were converted into an Agile backlog in Jira under the “My Software Team” project (project key: SCRUM). The backlog currently contains eight Epics and sixteen user stories. The stories are distributed across two completed Scrum sprints and are marked Done, providing traceability from the portal requirements to implementable work items.

**Current Jira structure:** 8 Epics, 16 user stories, story-point estimates, assignees, and sprint information. The Jira project contains the following backlog items.

| Jira Key | Epic                                 | User Story                 | Assignee        | Points | Sprint   |
|----------|--------------------------------------|----------------------------|-----------------|--------|----------|
| SCRUM-13 | User Authentication & Access Control | User Login                 | Chiranth J      | 3      | Sprint 1 |
| SCRUM-14 | User Authentication & Access Control | Role-Based Access          | Chiranth J      | 5      | Sprint 1 |
| SCRUM-15 | Event Proposal & Budget Management   | Create Event Proposal      | Chiranth J      | 5      | Sprint 1 |
| SCRUM-16 | Event Proposal & Budget Management   | Submit Event Proposal      | Chiranth J      | 3      | Sprint 1 |
| SCRUM-17 | Proposal Review & Budget Approval    | Review Event Proposal      | Chris John Paul | 5      | Sprint 1 |
| SCRUM-18 | Proposal Review & Budget Approval    | Approve or Reject Budget   | Chris John Paul | 5      | Sprint 1 |
| SCRUM-19 | QR Ticket Generation                 | Generate QR Tickets        | Chris John Paul | 8      | Sprint 1 |
| SCRUM-20 | QR Ticket Generation                 | Generate Unique Ticket IDs | Chris John Paul | 5      | Sprint 2 |
| SCRUM-21 | Event Ticket Booking                 | Browse Approved Events     | Deeksha G       | 3      | Sprint 2 |
| SCRUM-22 | Event Ticket Booking                 | Purchase Event Ticket      | Deeksha G       | 8      | Sprint 2 |
| SCRUM-23 | QR Ticket Access                     | View Purchased QR          | Deeksha G       | 3      | Sprint 2 |

|          |                               |                               |           |   |          |
|----------|-------------------------------|-------------------------------|-----------|---|----------|
|          |                               | Ticket                        |           |   |          |
| SCRUM-24 | QR Ticket Access              | Download QR Ticket            | Deeksha G | 2 | Sprint 2 |
| SCRUM-25 | Registration Monitoring       | View Event Registration       | Janhavi R | 3 | Sprint 2 |
| SCRUM-26 | Registration Monitoring       | View Ticket Availability      | Janhavi R | 3 | Sprint 2 |
| SCRUM-27 | System Security & Performance | Enforce Authenticated Access  | Janhavi R | 5 | Sprint 2 |
| SCRUM-28 | System Security & Performance | Maintain Response Performance | Janhavi R | 5 | Sprint 2 |

**Traceability to the system requirements.** The backlog covers authentication and role control, event proposal and budget submission, proposal review and budget decisions, QR ticket generation, event browsing and ticket purchase, ticket access, registration monitoring, and security/performance requirements. This aligns the Agile implementation plan with the functional and non-functional requirements used for the component model in this report.

**Backlog status.** All sixteen listed user stories are currently in the Done state in Jira. Sprint 1 and Sprint 2 are recorded as completed in the Jira project.